CURRICULUM VITAE: SOUMIK PURKAYASTHA

Primary appointment:
Assistant Professor (tenure-stream)
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Secondary appointment:
Mathematical Statistician
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1. Academic Interests

- Causal discovery and inference
- Spatiotemporal analysis/Infectious disease modeling
- Health services research

2. Education and Training

Year(s)	University, city, state	Degree, Year Awarded	Field
2014-17	St. Xavier's College (Autonomous), Kolkata, WB, INDIA	B.Sc. (Hons.), 2017	Statistics
2017-19	Indian Statistical Institute, Kolkata, WB, INDIA	M.Stat., 2019	Statistics
2019-21	University of Michigan, Ann Arbor, MI, USA	M.S., 2021	Biostatistics
2019-24	University of Michigan, Ann Arbor, MI, USA	Ph.D., 2024	Biostatistics

3. Appointments and Positions

Academic

Year(s)	Title	Department, School, Name and location of institution
2019-23	Graduate Student Research Assistant	Department of Biostatistics, University of Michigan, Ann Arbor, MI, USA.
2023-24	Rackham Predoctoral Fellow	Rackham Graduate School, University of Michigan, Ann Arbor, MI, USA.
2024-	Assistant Professor (tenure stream)	Department of Biostatistics and Health Data Science, University of Pittsburgh, Pittsburgh, PA, USA.
2024-	Mathematical Statistician	Center for Healthcare Evaluation, Research, and Promotion, VA Pittsburgh Healthcare System, Pittsburgh, PA, USA.

Non-Academic

Years	Title	Name and location of organization
2018	Data Science intern	Walmart Labs, Bengaluru, KA, INDIA
2021	AI/ML intern	Apple Inc., Cupertino, CA, USA

4. Membership in Professional and Scientific Societies

Years Inclusive	Name of Society
2021-	International Biometrics Society, Eastern North American Region (ENAR)
2021-	American Statistical Association (ASA)
2021-	Institute of Mathematical Statistics (IMS)
2022-	International Biometrics Society, Western North American Region (WNAR)
2023-	International Indian Statistical Association (IISA)
2024-	International Chinese Statistical Association (ICSA)
2026-	International Statistical Institute (ISI)

5. Honors

Year of Award	Title of Award and Awarding Association
2017-19	*Merit-based scholarship for good academic performance*, Indian Statistical Institute, Kolkata, WB, INDIA.
2019	*Sabyasachi Roy Memorial Gold Medal*, Indian Statistical Institute, Kolkata, WB, INDIA.
2019	*Michigan Data Science Challenge Winner*, Michigan Institute of Data Science, University of Michigan, Ann Arbor, MI, USA.
2020	*Richard G. Cornell Fellowship*, (awarded annually) Department of Biostatistics, University of Michigan, Ann Arbor, MI, USA.
2022-24	*Rackham Conference Travel Grant* (awarded annually), Rackham Graduate School, University of Michigan, Ann Arbor, MI, USA.
2023	*Rising Star Award*, School of Public Health, University of Michigan, Ann Arbor, MI, USA.
2023	*Best Paper Award* and *Best Presentation (Runner-up) Award*, Western North American Region of the International Biometrics Society.
2023-24	*Rackham Predoctoral Fellowship*, Rackham Graduate School, University of Michigan, Ann Arbor, MI, USA.

6. Professional Activities

6a. Teaching

Courses Taught

Year(s)	Course Number, Title	Credits, Average Enrollment	Role in course
2025 Fall	BIOST 2155: Introductory Statistical Learning for Health Sciences	2 credits, 14 students	Primary instructor

2026 Spring	BIOST 2025: Biostatistics Seminar	1 credit, 16 students	Primary instructor
2026 Fall	BIOST 2155: Introductory Statistical Learning for Health Sciences	2 credits, 12 students	Primary instructor
2026 Fall	PUBHLT 0411: Statistical Packages for Public Health Analysis	3 credits, 6 students	Co-instructor
2027 Spring	BIOST 2025: Biostatistics Seminar	1 credit, 16 students	Primary instructor

Other Teaching (lectures, tutorials, and continuing education courses)

Year(s)	Type of Teaching	Title
2024 Fall	Guest lecture	Fundamentals of Public Health, BS in Public Health Degree Program, School of Public Health, University of Pittsburgh
2024 Fall	Guest lecture	BIOST 2145 Health Data Science, MS in Biostatistics Degree Program, School of Public Health, University of Pittsburgh
2025 Fall	Guest lecture	Fundamentals of Public Health, BS in Public Health Degree Program, School of Public Health, University of Pittsburgh

Major Advisor for Graduate Student Essays, Theses, and Dissertations

Name of Student	Year of Degree	Type of Document	Notes
Zhiyi Yang	PhD in Biostatistics (202x expected)	Dissertation, name not fixed yet	Dissertation co-advisor
Feier Chang	PhD in Biostatistics (202x expected)	Dissertation, name not fixed yet	Dissertation advisor
Vani Patel	MS in Biostatistics (2026 expected)	Thesis, name not fixed yet	Thesis advisor

Service on Masters or Doctoral Committees

Name of Student	Degree Awarded Year	Type of Document
Joseph Armstrong	MS in Biostatistics (2026)	Thesis, *Prediction of PM 2.5 Using Geospatial Machine Learning Methods*
Sanika Kamath	MS in Human Genetics (2026)	Thesis, *Leveraging Single-Cell Reference Atlases to Enhance Predictive Biomarker Discovery for Cancer Immunotherapy*
Xin Li	MS in Biostatistics (2026)	Thesis, *Emerging Risk Factors and Temporal Patterns in Hospitalized COPD Patients, 2019-2023*
Crystal Zang	PhD in Biostatistics (2026)	Dissertation, *Supervised Dimensionality Reduction Methods for Large-Scale EHR and Omics Data in Biomedical Studies*

Service on Comprehensive or Qualifying Committees

Year(s)	Student Population	Type of Exam
2024-25	16 M.S. students	MS in Biostatistics Comprehensive Exam committee
2025-26	6 M.S. students	MS in Genome Bioinformatics Comprehensive Exam committee
2025-26	18 M.S. students	MS in Biostatistics Comprehensive Exam committee

Mentoring of Early and Mid-Career Faculty

Dates	Name of Faculty	Position of Faculty Member
2025- (Awarded)	Joshua Gordon (Competitive Career Development Fund, VA Pittsburgh Healthcare System)	Affiliate Investigator, Center for Healthcare Evaluation, Research, and Promotion, VA Pittsburgh Healthcare System
2026- (Presubmission)	Manisha Bhattacharyya (Career Development Award, VA Health Services Research)	COIN Investigator, Center for Healthcare Evaluation, Research, and Promotion, VA Pittsburgh Healthcare System

Academic Advising

Name of Student	Degree Awarded Year	Position	Year(s)
Alexandra Tirendi	MS in Biostatistics (2027 expected)	Academic Advisor	2025-26
Feier Chang	PhD in Biostatistics (202x expected)	Academic Advisor	2025-
Vani Patel	MS in Biostatistics (2026 expected)	Academic Advisor	2026-
Sharvil Sankriti	MS in Biostatistics (202x expected)	Academic Advisor	2026-

Supervision and Mentoring of Student/Graduate Student Researchers (GSR)

Name of Student	Degree Awarded Year	Position
Jipeng Zhang	PhD in Biostatistics (202x expected)	PhD GSR
Xizhi Wu	PhD in Rehabilitation Science (202x expected)	PhD GSR

Other Supervision and Mentoring

Dates	Name	Position
2024-	David Frank	Health Science Specialist, Center for Healthcare Evaluation, Research, and Promotion, VA Pittsburgh Healthcare System

6b. Research and Training

Grants and Contracts <u>Received</u>

Completed: Co-Investigator/Statistician

Year(s)	Grant and/or Contract Number and Title	Source	Award Amount	%Effort
2024-25	*Health-related Social Needs and Suicidality Among Sexual Minority Veterans.* P.I. J. Gordon	VA CCDF	$47,217	15% (1.8CM)

2026	*Development of Clinical Prediction Models for Pulmonary Outcomes in Sarcoidosis,* P.I.: M. Seedahmed (Diversity Supplement Award)	National Institutes of Health	$116,621	10% (1.2CM)

Current: Co-Investigator/Statistician

Year(s)	Grant and/or Contract Number and Title	Source	Award Amount	%Effort
2026-29	*I01 RD000382: Characterizing the Quality and Equity of Evidence-based Prescribing for Hospitalized Veterans.* P.I.: T. Anderson and U. Essien	VA Health Systems Research	$800,000	10% (1.2CM)
2024-27	*Developing User-Centered Reporting Strategies to Promote Health Equity in the Veterans Health Administration.* P.I.: L. R. M. Hausmann	VA Office of Health Outcomes	$2,470,354	25% (3.0CM)
2026-27	*Accelerating the Identification and Mitigation of Insulin Pump Risks using Artificial Intelligence,* P.I.: R. Codario and J. Estock	Breakthrough T1D	$189,038	15% (1.8CM)
2025-26	*MISP101143: HPV Vaccination and the Future Burden of HPV+ Oropharyngeal Cancer in the Veterans Health Administration.* *P.I.: J. Zevallos*	Merck	$334,094	15% (1.8CM)
2026-27	*The Impact of Service-related Burn Pit and Embedded Metal Fragments Exposure on Sarcoidosis Incidence and Mortality.* *P.I.: M. Seedahmed*	VA Airborne Hazards and Burn Pits Center of Excellence	$367,539	10% (1.2CM)

Grants and Contracts <u>Pending</u>

Year(s)	Grant and/or Contract Number and Title	Source	Award Amount	%Effort
2027-28 (**JIT**)	*I21 RD002895-01: Optimizing HPV Vaccination Strategies for Head and Neck Cancer Prevention in Veterans* P.I.: J. Zevallos	VA Health Systems Research	$230,000	10% (1.2CM)
2027-32 (**awarded**)	*IK2 RD002006-01A1: Understanding How Military Exposures Influence Sarcoidosis Patterns, Prognosis, and Treatment Response to Improve Outcomes for Veterans* P.I. M. Seedahmed	VA Health Systems Research (CDA-2)	$1,890,698	20% (2.4CM)
2027-28 (*proposed*)	*Safety Architecture for Veteran-use Electronic Medical Devices (SAVEMD)* P.I.s: J. Estock and R. Codario	VA Health Systems Research	$350,000	15% (1.8CM)

2027-28 (*proposed*)	*Resolving the Causal Direction of Depression and Inflammation: A Survey-Weighted Structure Learning Approach* P.I. S. Purkayastha	National Institutes of Health (R21)	$413,296	15% (1.8CM)
2027-28 (**JIT**)	*Evaluating the VA National Pharmacogenomic Program to Reduce Adverse Drug Events in Older Adults in the VA Nursing Home* P.I. C. Kistler	National Institutes of Health (R21)	$392,198	10% (1.2CM)
2027-32 (*proposed*)	*Advancing Evidence-Based Prescribing for Patients Hospitalized with Cardiopulmonary Conditions* *P.I.s: T. Anderson and U. Essien*	National Institutes of Health (R01)	$1,780,730	10% (1.2CM)

Invited Presentations Related to Research

Date	Title of Presentation	Venue
Jun 2025	Examining Directional Association between Depression and Anxiety	International Indian Statistical Association Conference, *Lincoln, USA*
Jul 2025	Examining Directional Association between Depression and Anxiety	New Researchers Conference, Institute of Mathematical Statistics, *Nashville, USA*
Dec 2025	A Mechanistic Framework for Collider Detection in Observational Data	Int. Conf. on Computational and Financial Econometrics and Computational and Methodological Statistics, *London, UK*
Dec 2025	A Mechanistic Framework for Collider Detection in Observational Data	Jt. Meetings of 2025 Taipei Int. Statistical Symposium and 13th ICSA Int. Conference, *Taipei, TAIWAN*
Jun 2026	Quantification and Inference of Asymmetric Relations Under Generative Exposure Mappings	ISI Regional Statistics Conference, *Valetta, MALTA*
Jul 2026	Causal Discovery for Self-Reported Mental Health Data	IBS International Biometric Conference, *Seoul, SOUTH KOREA*
Aug 2026	Causal Discovery for Self-Reported Mental Health Data	ASA Joint Statistical Meeting, *Boston, USA*
Dec 2026 (**upcoming**)	Survey-Weighted Ordinal Causal Discovery: Application to Veteran Social Needs and Health Indicators	IISA Annual Meeting, *Varanasi, INDIA*
Mar 2027 (**upcoming**)	Survey-Weighted Ordinal Causal Discovery: Application to Veteran Social Needs and Health Indicators	ENAR Annual Meeting, *Boston, USA*

Other Research and Training Activities

Date	Name	Venue
Mar 2026	2026 Program in Quantitative Genomics: AI for Medical Data	Harvard School of Public Health, *Boston, USA*

6c. Publications; *h-index: 11 (Google scholar) as of October 2026; +: co-first author; *: trainee., #: conference proceedings.*

Refereed Articles – Published or In Press (most-recent first)

2026 (while at the University of Pittsburgh/VA Pittsburgh Healthcare System)

1. Kadura, S., Purkayastha, S., Benditt, J., Anand, A., Collins, B., De Quadros, M., Hobson, M., Biswas, M.J., Ho, L., Spino, C., and Raghu, G., (2026) *Yoga Effect on Quality-of-Life Study Among Patients with Idiopathic Pulmonary Fibrosis (YES-IPF)*. DOI 10.1016/j.rmed.2026.109194
2. Sethi, V., Gunabushanam, V., Liu, H., Purkayastha, S., Bardhi, E., Geller, D., Ganesh, S., Hughes, C., and Humar, A., (2026), *Outcomes After Living Donor Liver Transplant for Intrahepatic Cholangiocarcinoma*. American Journal of Transplantation, DOI 10.1016/j.ajt.2026.09.039
3. Helou, V., Carlson, K. M., **Purkayastha, S.**, Mazul, A. L., Contrera, K. J., and Zevallos, J. P. (2026) *Evolving Trends in Definitive Surgical Management of Oropharyngeal Squamous Cell Carcinoma in the United States*, Head & Neck: 1–9, DOI 10.1002/hed.70396
4. Sethi, V., **Purkayastha, S.**, Liu, H., Spitz, F., Abdullah, A., Demirors, B., Cruz Jr, R.J., Hariharan, S., Puttarajappa, C. and Molinari, M., (2026). *Drug Overdose Mortality Decline and Deceased Donor Transplantation in the United States*. Am. Journal of Transplantation. DOI: 10.1016/j.ajt.2026.05.715.
5. # Ren, J., A. Parr, S**. Purkayastha**, M. Mor, K. Brown, J. J. Tang, L. Hausmann, W. Gellad, U. R. Essien, and T. S. Anderson., (2026) *Variation in Guideline-Concordant Inhaler Regimen Intensification Following Chronic Obstructive Pulmonary Disease Hospitalizations Across Veterans Affairs Medical Centers*. American Journal of Respiratory and Critical Care Medicine 212, no. Supplement 1: aamag162-923.
6. Anderson, T. S., **Purkayastha, S.**, Parr, A., Berstein, E. Y., Bachrach, R. L., Mor, M. K., Gellad, W. F., Hausmann, L. R. M., Fine, M. J., Essien, U. R. (2026). *Initiation of Medications for Alcohol Use Disorder Among Hospitalized Veterans: A Retrospective Cohort Study*. Annals of Internal Medicine, DOI: 10.7326/annals-26-00089.
7. **Purkayastha, S.**, Basu, A. (2026) *On Minimum Bregman Divergence Inference*. Mathematics, 14(4):670.
8. # Sethi, V., **Purkayastha, S.**, Liu, H., Spitz, F., Hariharan, S., Puttarajappa, C., & Molinari, M. (2026). *Drug Overdose Mortality Decline and Kidney Transplantation in the United States*. American Journal of Transplantation, 26(1), S94-S95.
9. Russell, L. E.+, Frank, D. A.+, **Purkayastha, S.**, McCoy, J. L., Leder, S. M., Gordon, J. H., Lamba, S., Procario, G. T., Moy, E. M., Hausmann, L. R. M. (2026), *Racial, Ethnic, and Sex Differences in Social Risks and Social Needs Concordance Among Veterans*. JAMA Network Open, 9(2), e2559892.
10. **Purkayastha, S.** and Song, P. X. K., (2026), *Quantification and cross-fitting inference of asymmetric relations under generative exposure mapping models*. Statistica Sinica, DOI: 10.5705/ss.202025.0236.

2025 (while at the University of Pittsburgh/VA Pittsburgh Healthcare System)

11. # Bessette, L., Essien, U., Parr, A., Mor, M., **Purkayastha, S.**, Gellad, W., Hausmann, L., Fine, M. and Anderson, T., (2025). *Underutilization of guideline-directed medical therapy following hospitalizations for heart failure with reduced ejection fraction among US Veterans, 2022 to 2023*. Circulation, 152(3), A4359558.
12. Lamba, S., Frank, D., McCoy, J.L., Russell, L.E., **Purkayastha. S.**, Gordon, J.H., Leder, S.M., Procario, G.T., Moy, E.M., Hausmann, L.R.M., (2025), *Health-Related Social Needs Among LGB+ Veterans*. JAMA Network Open, 8(10), e2539986
13. Zhang, L., **Purkayastha, S**., Kirsner, R., Spino, C. and Song, P. X. K., (2025) *Determinants of Enrolment in 284 Clinical Trials for Healing Diabetic Foot Ulcers: A Systematic Review*. BMJ Open, 15(7), e095512.

2020-2024 (while at the University of Michigan)

14. **Purkayastha, S.** and Song, P.X.K., (2024). *fastMI: A fast and consistent copula-based nonparametric estimator of mutual information.* Journal of Multivariate Analysis, 201, p.105270.
15. Salvatore, M.+, **Purkayastha, S.+**, Ganapathi, L., Bhattacharyya, R., Kundu, R., Zimmermann, L., Ray, D., Hazra, A., Kleinsasser, M., Solomon, S. and Subbaraman, R., (2022). Lessons from SARS-CoV-2 in India: A data-driven framework for pandemic resilience. Science Advances, 8(24), p.eabp8621.
16. Bhaduri, R.+, Kundu, R.+, **Purkayastha, S.**, Kleinsasser, M., Beesley, L.J., Mukherjee, B. and Datta, J., (2022). *Extending the susceptible-exposed-infected-removed (SEIR) model to handle the false negative rate and symptom-based administration of COVID-19 diagnostic tests: SEIR-fansy*. Statistics in Medicine, 41(13), pp.2317-2337.
17. Zimmermann, L., Bhattacharya, S., **Purkayastha, S.**, Kundu, R., Bhaduri, R., Ghosh, P. and Mukherjee, B., (2021). *SARS-CoV-2 infection fatality rates in India: systematic review, meta-analysis and model-based estimation*. Studies in Microeconomics, 9(2), pp.137-179.
18. **Purkayastha, S.,** Kundu, R., Bhaduri, R., Barker, D., Kleinsasser, M., Ray, D. and Mukherjee, B., (2021). *Estimating the wave 1 and wave 2 infection fatality rates from SARS-CoV-2 in India.* BMC Research Notes, 14, pp.1-7.
19. **Purkayastha, S.**, Bhattacharyya, R., Bhaduri, R., Kundu, R., Gu, X., Salvatore, M., Ray, D., Mishra, S. and Mukherjee, B., 2021. *A comparison of five epidemiological models for transmission of SARS-CoV-2 in India.* BMC Infectious Diseases, 21, pp.1-23.
20. Salvatore, M., Basu, D., Ray, D., Kleinsasser, M., **Purkayastha, S.**, Bhattacharyya, R. and Mukherjee, B., (2020). *Comprehensive public health evaluation of lockdown as a non-pharmaceutical intervention on COVID-19 spread in India: national trends masking state-level variations*. BMJ open, 10(12), p.e041778.
21. **Purkayastha, S.** and Song, P., (2021). *Discussion on "The timing and effectiveness of implementing mild interventions of COVID-19 in large industrial regions via a synthetic control method" by Tian et al.* Statistics and Its Interface, 14(1), pp.21-22.
22. Tang, L., Zhou, Y., Wang, L., **Purkayastha, S.**, Zhang, L., He, J., Wang, F. and Song, P.X.K., (2020). *A review of multi-compartment infectious disease models.* International Statistical Review, 88(2), pp.462-513.
23. **Purkayastha, S.**, Salvatore, M. and Mukherjee, B., (2020). *Are women leaders significantly better at controlling the contagion during the COVID-19 pandemic?* Journal of Health and Social Sciences, 5(2), p.231.
24. Zhou, Y., Wang, L., Zhang, L., Shi, L., Yang, K., He, J., Bangyao, Z., Overton, W., **Purkayastha, S.**, & Song, P. (2020). *A Spatiotemporal Epidemiological Prediction Model to Inform County-Level COVID-19 Risk in the United States*. Harvard Data Science Review, (Special Issue 1)
25. Giri, S., **Purkayastha, S.**, Hazra, S., Chanda, A., Das, I. and Das, S., (2020). *Prediction of monthly Hilsa (Tenualosa ilisha) catch in the Northern Bay of Bengal using Bayesian structural time series model.* Regional Studies in Marine Science, 39, p.101456.
26. Ray, D., Salvatore, M., Bhattacharyya, R., Wang, L., Du, J., Mohammed, S., **Purkayastha, S.**, Halder, A., Rix, A., Barker, D., Kleinsasser, M., Zhou, Y., Bose, D., Song, P., Banerjee, M., Baladandayuthapani, V., Ghosh, P., & Mukherjee, B. (2020). *Predictions, Role of Interventions, and Effects of a Historic National Lockdown in India's Response to the COVID-19 Pandemic: Data Science Call to Arms*. Harvard Data Science Review, (Special Issue 1).

Refereed Articles – Submitted/Preprint

1. **Purkayastha, S.** and Song, P. X. K., (2026), *A Framework for Covariate-Adjusted Bivariate Causal Discovery.* PREPRINT UPON REQUEST,
2. **Purkayastha. S.,** Russell, L.E., Frank, D., Hausmann, L.R.M. (2026) *Survey-Weighted Ordinal Causal Discovery: Application to Veteran Social Needs, Mental Health, and Well-Being*. [Paper and supplement]
3. Chang, F.*, and **Purkayastha, S. (2026)**. *Fed-GEE: Federated generalized estimating equations for privacy-preserving population-level inference for multicenter longitudinal data* [Paper and supplement].
4. Bernstein, E. Y., Bessette, L. G., **Purkayastha, S.**, Parr, A., Mor, M. K., Park, T. W., Hausmann, L. R. M., Gellad, W. F., Essien, U. R., & Anderson, T. S. (2026). *Post-hospitalization persistence to medications for alcohol use disorder.* PREPRINT UPON REQUEST
5. Gambino, S., Wu, X., Codario, R., Wang, Y., **Purkayastha, S.,** and Estock, J., (2026), *Natural language processing to extract insulin pump risks from adverse event narratives.* PREPRINT UPON REQUEST
6. Li, X., Liu, J., **Purkayastha, S.**, Talisa, V. B., Rosser, F. J., and Ding, Y. (2026). *Emerging Risk Factors and Temporal Patterns in Hospitalized Patients With COPD: A Retrospective Analysis of the NIS Data (2019–2023)*. PREPRINT UPON REQUEST
7. Bessette, L. G., Purkayastha, S., Parr, A., Mor, M. K., Hausmann, L. R. M., Gellad, W. F., Essien, U. R., & Anderson, T. S. (2026). *Hospital-level variation in discharge prescribing of heart failure guideline-directed medical therapy*. PREPRINT UPON REQUEST

Published Abstracts (most-recent first)

1. Helou, V., Carlson, K. M., **Purkayastha, S.**, Mazul, A. L., Contrera, K. J., Zevallos, J. P. (2026) *Evolving Trends in Definitive Surgical Management of Oropharyngeal Squamous Cell Carcinoma: A National Cancer Database Analysis*. American Head and Neck Society 2026 Annual Meeting, Boston, USA.
2. Bernstein, E. Y., Bessette, L. G., Parr, A., **Purkayastha, S.**, Mor, M. K., Gellad, W. F., Hausmann, L. R. M., Essien, U. R., Anderson, T. S., (2026) *Persistence to medications for alcohol use disorder initiated during hospitalization.* 2026 Annual Meeting of the Society of General Internal Medicine, Washington D.C., USA.
3. Bessette, L. G., Essien, U. R., Parr, A., Mor, M. K., **Purkayastha, S.**, Gellad, W. F., Hausmann, L. R. M., Anderson, T. S., (2026) *Adherence and persistence to SGLT2 inhibitors following hospitalizations for heart failure among U.S. Veterans.* 2026 Annual Meeting of the Society of General Internal Medicine, Washington D.C., USA.
4. Essien, U.R., **Purkayastha, S.**, Parr, A., Mor, M. K., Tang, J. J., Gellad, W. F., Hausmann, L. R. M., Anderson, T. S., (2026) *Hospital-Level Variation in Quality and Equity of Prescribing Medications for Alcohol Use Disorder in VA*. 2026 Annual Meeting of the Society of General Internal Medicine, Washington D.C., USA.
5. Sethi, V., **Purkayastha, S.**, Liu, H., Zhang, X. M., Abdullah, A., Spitz, F., Elias, C., Demirors, B., Puttarajappa, C., Molinari, M., (2025) *Impact of Drug Overdose Mortality Decline on Kidney Transplantation in the US.* 21st Annual Academic Surgical Congress, Orlando, USA.
6. Hausmann, L. R. M., Russell, L. E., Frank, D. A., **Purkayastha, S.**, McCoy, J. L., Leder, S. M., Gordon, J. H., Procario, G. T., Lamba, S., Moy, E. M., (2025) *Racial, Ethnic, and Sex Differences in Agreement between Self-reported Social Risks and Need for Support*. Society of Behavioral Medicine 47th Annual Meeting & Scientific Sessions, Chicago, USA.
7. Lamba, S., Frank, D., McCoy, J.L., Russell, L.E., **Purkayastha. S.**, Gordon, J.H., Leder, S.M., Procario, G.T., Moy, E.M., Hausmann, L.R.M., (2025). *The Prevalence of Social Needs Among LGB+ Veterans: A National Survey of VHA Primary Care Patients*. 2025 Academy Health Annual Research Meeting, Minneapolis, USA.

6d. Presentations (most-recent first)

1. ENAR Annual Meeting (Mar 2027). "*Survey-Weighted Ordinal Causal Discovery: Application to Veteran Social Needs and Health Indicators*", Boston, USA (**upcoming**)
2. IISA Annual Meeting (Dec 2026). "*Survey-Weighted Ordinal Causal Discovery: Application to Veteran Social Needs and Health Indicators*", Varanasi, INDIA (**upcoming**)
3. ASA Joint Statistical Meeting (August 2026). "*Causal Discovery for Self-Reported Mental Health Data*". Boston, USA
4. IBS International Biometric Conference (July 2026). "*Causal Discovery for Self-Reported Mental Health Data*". Seoul, SOUTH KOREA
5. ISI Regional Statistics Conference (June 2026). "*Causal Discovery for Self-Reported Mental Health Data*". Valetta, MALTA
6. Joint Meetings of 2025 Taipei International Statistical Symposium and 13th ICSA International Conference (December 2025), "*A Mechanistic Framework for Collider Detection in Observational Data*". Taipei, TAIWAN.
7. 19th International Joint Conference on Computational and Financial Econometrics (CFE) and Computational and Methodological Statistics: CMStatistics (December 2025). *"A Mechanistic Framework for Collider Detection in Observational Data".* London, UK.
8. 8th International Conference on Econometrics and Statistics: EcoSta (August 2025). *"Examining Directional Association between Depression and Anxiety".* Tokyo, JAPAN.
9. Joint Statistical Meetings (August 2025). *"Examining Directional Association between Depression and Anxiety".* Nashville, USA.
10. International Indian Statistical Association Conference (June 2025). "*Examining Directional Association between Depression and Anxiety".* Lincoln, USA.
11. Eastern North American Region (ENAR) of the International Biometric Society Spring Meeting (March 2025). "*Examining Directional Association between Depression and Anxiety".* New Orleans, USA.
12. Joint Statistical Meetings (August 2023). *"Asymmetric predictability in causal discovery: an information theoretic approach".* Toronto, CANADA.
13. Western North American Region (WNAR) of the International Biometric Society Spring Meeting (June 2023). *"Asymmetric predictability in causal discovery: an information theoretic approach".* Anchorage, USA.
14. Eastern North American Region (ENAR) of the International Biometric Society Spring Meeting (March 2023). *"An information-theoretic framework for causal discovery in epigenetic data".* Nashville, USA.

6e. Non-Print Media (Software, electronic)

1. Bhaduri R.[+], Kundu R.[+], **Purkayastha S.**, Beesley L., Mukherjee B., Kleinsasser, M. (2021). *SEIRfansy: Extended Susceptible-Exposed-Infected-Recovery Model* [link]
2. **Purkayastha, S.** and Song, P.X.K., 2024. *fastMI: A fast and consistent copula-based nonparametric estimator of mutual information.* [link]
3. **Purkayastha, S.** and Song, P.X.K., 2024. *comet: Collider-mediator testing using information theory* [link]
4. **Purkayastha, S.** and Song, P.X.K., 2026: *causalGEM: Bivariate causal discovery via the Generative Exposure Model* [link]
5. **Purkayastha. S.,** Frank, D., Russell, L.E., Hausmann, L.R.M., 2026: *swaoBN: Survey-Weighted, Covariate-Adjusted Ordinal Bayesian Network Learning* [link]

6. Chang, F.* and **Purkayastha, S.**: *FedGEE: Federated Generalized Estimating Equations for Privacy-Preserving Population-Level Inference for Multicenter Longitudinal Data* [link]

6f. Technical and Government Reports

1. *Opportunities for Equity-Guided Improvement in GLP-1 Agonist Prescribing in VA,* VA Office of Health Equity, P.I. Carolyn Thorpe.

6g. Other Publications/Press Coverage

1. Gotkine, E., Medications initiated in 30% of hospitalizations for alcohol use disorder among Veterans, MedicalXpress, 2026
2. Bastian, H., Women Versus Men Leaders in the Pandemic: An Update and Dig Into the Latest Data. PLOS Blogs, 2022
3. Laguipo, A., A study of the COVID fatality rates in India during waves 1 and 2. News-Medical.Net, 2021.
4. Mukherjee, B., **Purkayastha, S.**, Salvatore, M., & Mishra, S. Underreporting does hurt the COVID fight. The Hindu, 2021.
5. Basu, D., Salvatore, M., Kleinsasser, M., **Purkayastha, S.,** Bhattacharya, R., and Mukherjee, B., We're Focusing on National Data on COVID-19 When We Should Be Looking at State- Level Trends. The Wire, 2020.

7. Service

7a.i. Service to Department

Year(s)	Activity
2024-25	Graduate Research and Development Seminar Organizer
Mar 2025	Research Day Poster Competition Judge
Mar 2026	Research Day Poster Competition Judge
2025-27	Seminar and Research Day organizer
2025-26	PhD in Biostatistics Admissions Committee (300+ applications)

7a.ii. Service to School

Year(s)	Activity
Apr 2025	Dean's Day Poster Competition Judge
Apr 2026	Dean's Day Poster Competition Judge
2025-	Union Steward, Union of Pitt Faculty (United Steelworkers Local 1511)

7a.iii. Service to University

Year(s)	Activity
2026-27	University of Pittsburgh Health Sciences PhD Core Curriculum Committee (AI Utilization group)

7b. Manuscript and other Documentation/Publication Review

ASA Discoveries, Annals of Applied Statistics, Biometrics, BMC Medical Research Methodology, BMC Health Services Research, Communications in Statistics (Case Studies and Data Analysis), Health Services Research, Journal of the American Statistical Association, Journal of the Royal Statistical Society, Series A (Statistics in

Society), New England Journal of Statistics in Data Science, PLOS One, Scientific Reports, Social Psychiatry and Psychiatric Epidemiology, Statistics in Medicine, Statistical Science.

7c. Service to Field of Scholarship

Date(s)	Organization/Agency	Position and/or Type of Service
Dec 2025	American Statistical Association, Health Policy in Statistics Section	Student Paper Competition Reviewer

7d. Leadership in Scholarly and Professional Organizations and Honorary Societies

Date(s)	Organization/Agency	Position and/or Type of Service
2026-	American Statistical Association, Health Policy in Statistics Section	Secretary

7e. Governmental Organizations

Date(s)	Organization/Agency	Position and/or Type of Service
Jan 2025	Congressionally Directed Medical Research Program (CDMRP) Discovery Award, US Department of Defense	Specialist Biostatistician Reviewer
2025-26	Center for Healthcare Evaluation, Research, and Promotion, VA Pittsburgh Healthcare System	Internal Biostatistician Reviewer